`Min-p`, Max Exaggeration: A Critical Analysis of `Min-p` Sampling in Language Models

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Abstract

Sampling from language models impacts the quality and diversity of generated outputs, affecting both research and real-world applications. Recently, Nguyen et al. (2024)'s "Turning Up the Heat: Min-p Sampling for Creative and Coherent LLM Outputs" introduced a new sampler called `min-p`, claiming it achieves superior quality and diversity over established methods such as `basic`, `top-k`, and `top-p` sampling. The paper was the 18th highest-scoring submission to ICLR 2025 and was selected for an Oral presentation. This paper conducts a comprehensive re-examination of the evidence supporting `min-p` and reaches different conclusions across the original paper's four main lines of evidence. First, the original paper's human evaluations omitted collected data for one baseline sampler, used a pooled statistical test that does not match the per-setting claim, and summarized qualitative feedback in a way the responses do not support; our reanalysis demonstrates `min-p` did not outperform baseline samplers in quality, diversity, or a trade-off between quality and diversity. In response to our findings, the authors of the original paper conducted a new human evaluation using a different sampler implementation, task, and rubric that nevertheless provides further evidence `min-p` does not improve over baseline samplers. Second, comprehensively sweeping the original paper's NLP benchmarks (GSM8K CoT and GPQA) reveals `min-p` does not surpass baseline samplers when controlling for the number of hyperparameters. Third, the original paper's LLM-as-a-Judge evaluations lack methodological clarity and appear inconsistently reported: the higher of two scores was reported for `min-p` while the lower of two scores was reported for `top-p`. Fourth, community adoption claims (49k GitHub repositories, 1.1M GitHub stars) were found to be unsubstantiated, leading to their removal from the ICLR 2025 Camera Ready; the revised adoption statement attributes the usage of the frameworks that integrate `min-p` to `min-p` itself. We conclude that evidence presented in the original paper fails to support claims that `min-p` improves quality, diversity, or a trade-off between quality and diversity.

1 Introduction

Large language model (LLM) capabilities have transformed numerous domains, from creative writing to scientific research. A critical detail of LLM deployment is the *sampling method*: the algorithm that determines how tokens are sampled during generation. Sampling strategies directly impact the quality and diversity of generated outputs, making them important to both research and deployment.

Commonly used samplers include `basic` (temperature-only) `sampling` (Ackley et al., 1985)[1], which samples tokens based on their temperature-scaled softmax-normalized logits; `top-k sampling` (Fan et al., 2018), which samples the k most probable tokens; and `top-p sampling` (Holtzman et al., 2020), which samples tokens comprising the top p probability mass. Other samplers include η`-sampling`, ϵ`-sampling` (Hewitt et al., 2022) and `mirostat sampling` (Basu et al., 2020).

[1]We cite Ackley et al. (1985) because, to our knowledge, it is the earliest work to sample discrete states from a temperature-scaled Boltzmann (softmax) distribution, which is exactly what temperature-only sampling does with a language model's logits.

Recently, the paper "Turning Up the Heat: Min-P Sampling for Creative and Coherent LLM Outputs" (Nguyen et al., 2024) introduced a new sampling method called `min-p sampling`, claiming it produces higher quality and higher diversity outputs than other samplers. The paper's influence extends well beyond its venue. Within a year of its release it accrued more than 170 citations, and `min-p` was integrated into most widely used open-source inference libraries, including Hugging Face Transformers, vLLM, SGLang, llama.cpp and Text Generation Inference, on the strength of its reported results. Its claims have also shaped subsequent research at top venues. Jiang et al. (2025) (NeurIPS 2025 Best Paper) used `min-p` as their representative decoding-time intervention against output homogeneity and, upon finding it insufficient, concluded that decoding-time interventions in general cannot preserve diversity; that generalization is only warranted if `min-p` is in fact a strong diversity-promoting sampler. Tan et al. (2026) (ICLR 2026 Oral) adopted `min-p` as the primary baseline for a new sampler and evaluated it with the same methodology whose flaws we document here. Whether `min-p`'s claimed advantages are real therefore matters not only to practitioners choosing a sampler, but to research that builds on those claims.

Given the potential impact of an improved sampling method and the paper's exposure as the 18th highest-scoring submission at ICLR 2025[2], we carefully scrutinized the methodologies, data, analyses, code and conclusions presented in support of `min-p` across the authors' four lines of evidence: (1) human evaluations, (2) natural language processing (NLP) benchmark evaluations, (3) LLM-As-A-Judge evaluations and (4) community adoption metrics. Our re-analyses of the evidence lead us to conclude that **relative to commonly used samplers, `min-p` does not improve quality or diversity or the trade-off between quality and diversity**. Our code and W&B sweeps of NLP benchmark evaluations will be publicly released and archived (e.g., via the Software Heritage Archive) upon acceptance and de-anonymization; we omit direct links here to preserve anonymity for the double-blind review process.

2 Re-Analyzing `Min-p`'s Human Evaluations

We began with re-analyzing the original paper's human evaluations since human judgments are widely considered the gold standard for assessing language model outputs (Van Der Lee et al., 2019; Roller et al., 2020; Howcroft et al., 2020; Clark et al., 2021; Liang et al., 2022; Khashabi et al., 2022; Chiang et al., 2024; Biderman et al., 2024; Schaeffer et al., 2025b). We identified four key issues.

2.1 Human evaluators' scores for one of two baseline samplers were omitted

Section 6 of Nguyen et al. (2024) states human participants evaluated `min-p` against a single sampler: `top-p`. Both the Oct 2024 Arxiv manuscript and ICLR OpenReview manuscript repeatedly state that `min-p` and `top-p` were considered, and their Table 4 presents results only for these. However, when examining the paper's data, we found that **scores for a second baseline sampler (`basic` sampling) were not included in the methodology, the analysis or the results, and the omission was not mentioned**. The authors publicly confirmed this. These omitted scores comprised $1/3^{\text{rd}}$ of the total collected scores. After we raised the issue, the omitted data were added to the Camera Ready's Table 4, but the methodology, the results and the conclusions have not been correspondingly updated.

2.2 Visualizations and Statistical Tests Fail to Support Claim That `Min-p` Outperforms Other Samplers

Based on the human evaluators' scores, Section 6 of Nguyen et al. (2024) concluded that "`min-p` sampling consistently scored higher than `top-p` sampling across all settings" and that "[a] paired t-test confirmed that the differences in scores between min-p and top-p sampling were statistically significant ($p < 0.05$)." However, **both visualizations and statistical hypothesis tests of the original human evaluation data suggest `min-p` is indistinguishable from the baselines in almost all settings.**

To briefly explain the human evaluation methodology, three samplers (`basic`, `top-p` and `min-p`) were compared in six conditions: three temperatures $(1.0, 2.0, 3.0)$ and two diversity settings ("high" and "low") corresponding to different p hyperparameters. Humans were asked to score the generated outputs under

[2] `https://papercopilot.com/statistics/iclr-statistics/iclr-2025-statistics/`

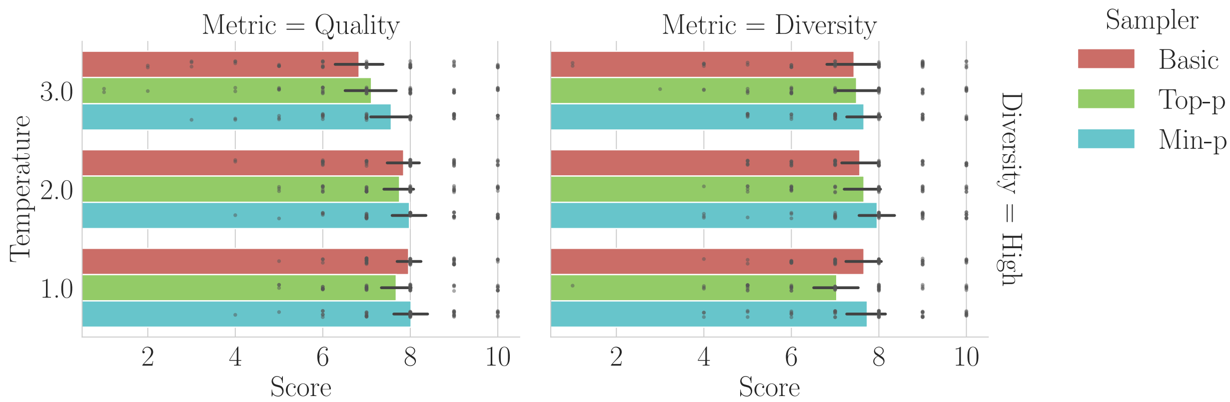

Figure 1: **Visualizing Human Evaluators' Scores from Nguyen et al. (2024)'s Data Demonstrates `Min-p` Does Not "Consistently" Outperform Other Samplers.** Rather, the original paper's data suggest `min-p` is largely indistinguishable from other samplers based on 95% confidence intervals.

Metric	Alt. Hyp.	Temperature					
		$\tau = 1.0$		$\tau = 2.0$		$\tau = 3.0$	
		t	p	t	p	t	p
Quality	`Min-p` > `Basic`	0.33	.370	0.65	.260	$3.13^{***\dagger}$	.001
	`Min-p` > `Top-p`	2.05^{*}	.023	1.18	.121	2.02^{*}	.025
Diversity	`Min-p` > `Basic`	0.31	.378	1.86^{*}	.034	0.85	.201
	`Min-p` > `Top-p`	2.64^{**}	.006	1.44	.078	0.87	.195

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, † Significant after Bonferroni correction for 12 comparisons.

Note: All tests were paired t-tests with df = 52, one-sided (alternative = "greater")

Table 1: **Hypothesis Testing of Human Evaluators' Scores Fails to Support Claim that `Min-p` Consistently Outperforms Other Samplers.** To test whether evidence supports the claim that `min-p` "consistently outperforms" other samplers, we conducted one-sided paired t-tests using the authors' published data. Without correcting for multiple comparisons, evidence exists to support `min-p`'s superiority in 5 of 12 comparisons at $\alpha = 0.05$ and 2 of 12 comparisons at $\alpha = 0.01$. After applying a Bonferroni correction for multiple comparisons, evidence exists to support `min-p`'s superiority in 1 of 12 comparisons at $\alpha = 0.05$ (quality, `min-p` > `basic`, at $\tau = 3.0$, the temperature at which all samplers receive their lowest quality scores) and 0 of 12 comparisons at $\alpha = 0.01$. For details, see Sec. 2.3.

two metrics: quality and diversity. Participants were excluded if they failed attention checks. For more information, please see the original manuscript.

We focused on the "high" diversity setting for three reasons: First, the claimed advantage of `min-p` sampling is that it provides both high quality and high diversity, whereas other samplers typically trade one off against the other. Second, the authors publicly recommended focusing on the high diversity setting, writing that "the low [diversity] settings were quite experimental". Third, `top-p`'s p value in the low diversity setting was atypically small ($p = 0.1$); after we raised this, the authors ran a new human evaluation that changed the low diversity `top-p` p from 0.1 to 0.9. We return to this second new human evaluation in Sec. 2.4.

We began by visualizing the human evaluations' scores from Nguyen et al. (2024). **Using the original paper's data, Fig. 1 reveals that the three samplers provide similar quality and similar diversity, with 95% confidence intervals frequently overlapping**.

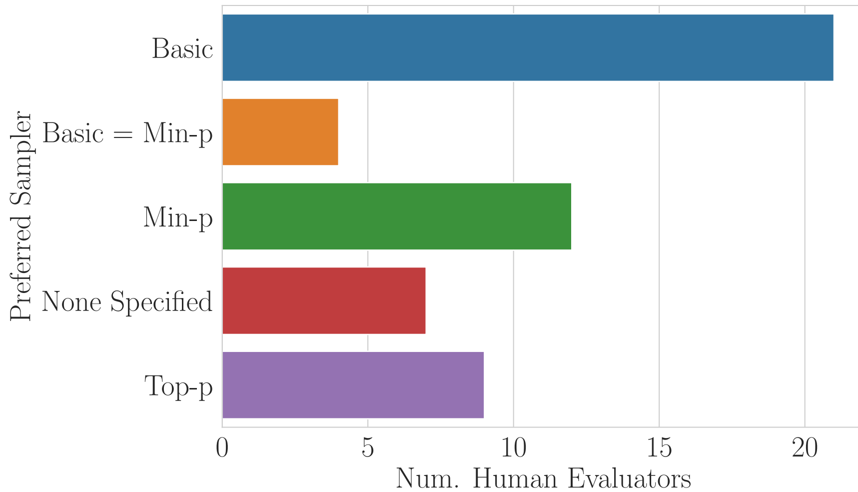

Figure 2: **Manual Annotation of Human Evaluators' Qualitative Responses Fail to Support Claim that `Min-P` Was the Preferred Sampler.** We manually annotated responses from human annotators regarding their preferred sampler(s) at the end of the original paper's study. The responses suggest `min-p` was not the most preferred sampler. We provide example responses in Sec. 2.3.

To more rigorously assess the claim that `min-p` consistently outperforms other samplers, we conducted 12 one-sided paired t-tests for each metric (quality or diversity), temperature $(1.0, 2.0, 3.0)$ and baseline sampler (`min-p` versus `basic`, `min-p` versus `top-p`). In each test, the null hypothesis is `min-p`'s score is less than or equal to the other sampler's score, and the alternative hypothesis is `min-p`'s score is greater than the other sampler's score. Statistical test results are displayed in Table 1. Without correcting for multiple comparisons, we found evidence to reject the null hypotheses in 5 of 12 tests at $\alpha = 0.05$ and 2 of 12 tests at $\alpha = 0.01$. After applying a Bonferroni correction for multiple comparisons, we found evidence to reject the null hypothesis in 1 of 12 tests at $\alpha = 0.05$ and 0 of 12 tests at $\alpha = 0.01$. **Based on the original paper's data, there is insufficient evidence to support the claim that `min-p` consistently outperforms baseline samplers across all settings.**

Furthermore, given that the original paper claims that `min-p` "consistently" scores higher, an Intersection-Union Test (IUT) may be the appropriate statistical test, where the alternative hypothesis is that `min-p` is better in all 12 comparisons and the null hypothesis is the set complement. Since the largest p-value of the 12 comparisons is 0.378, under the IUT, we again find insufficient evidence to reject the null hypothesis at both $\alpha = 0.05$ and $\alpha = 0.01$.

The original paper's statistical analysis reached a different conclusion for two reasons. First, despite claiming that `min-p` "consistently scored higher" "across all settings" (metric, temperature, and diversity), the paper pooled data across all settings and performed a single t-test, which tests whether `min-p` scored higher on average. Second, pooling over all settings folds in the "low" diversity condition, in which `top-p`'s atypically small $p = 0.1$ substantially lowered `top-p`'s scores; the authors themselves publicly described the low diversity settings as "quite experimental" and subsequently changed p in their new human experiment (Sec. 2.4). The pooled test therefore does not support the per-setting claim made in the text.

2.3 Human Evaluators' Qualitative Responses Fail to Support Claim That `Min-p` Is Preferred Over Other Samplers

At the end of the human evaluation study, the original paper asked human participants to qualitatively describe which sampler(s) they preferred. The paper stated that "[p]articipants frequently noted that outputs generated with min-p sampling were more coherent and creative, especially at higher temperatures." However, when reading through the paper's data, we believe that **the qualitative responses suggest a different preference pattern**. We manually annotated the qualitative responses and visualized our annotations of the humans' expressed preferences (Fig. 2), and publicly posted our annotations in the same format as the original paper. We found two results: (1) more human evaluators explicitly preferred `basic` sampling than preferred `min-p` sampling, and (2) `min-p` was only slightly preferred over `top-p`. We provide quotations from human evaluators favoring `basic` sampling in Appendix A.

2.4 New Human Evaluation Study Shows `Min-p` Does Not Outperform Baselines in Quality, in Diversity, or in a Tradeoff Between Quality and Diversity

In response to our feedback, the authors conducted and added a new human evaluation study to Appendix C.2. Their new study made multiple methodological changes:

- Different sampler implementation: switched from applying temperature *after* truncation to applying temperature *before* truncation.
- Different distribution of human participants from Prolific.
- Different sampling hyperparameters for `top-p`: switched from 0.1 and 0.9 to 0.9 and 0.95.
- Different sampling hyperparameters for `min-p`: switched from 0.2 and 0.05 to 0.1 and 0.05.
- Different allotted reading time: increased from 30 minutes to 45 minutes.
- Different sampled text: 3 short paragraphs were replaced with a single complete story.
- Different rubric for human participants to evaluate sampled outputs.

Regarding the new human evaluation data and results, we share two discoveries here: First, we believe one value is incorrectly reported: in Nguyen et al. (2024)'s Table 15, the average score of `min-p` at $p = 0.05$ and temperature $T = 2$ is reported as 7.80, but based on the authors' publicly posted data, we believe the correct numerical value should be 5.80. Second, more generally, the data show again that `Min-p` does not outperform baselines in quality, in diversity or in a favorable tradeoff between quality and diversity. In this new study, whenever `min-p` outperforms other samplers, it does so under conditions that yield lower absolute scores than other conditions (Fig. 3). For instance, `min-p` shows an advantage over the baselines in the "high" diversity setting at $T = 2$ and in the "low" diversity setting at $T = 3$. However, in both of these conditions, `min-p` receives lower quality and diversity scores than it does in the "high" diversity setting at $T = 1$ and the "low" diversity setting at $T = 2$. This shows `min-p`'s advantage is observed primarily under conditions that yield lower overall quality and diversity scores compared to other achievable conditions. **For anyone seeking higher quality or diversity, `min-p` offers no apparent advantage over `basic` or `top-p` sampling**. Reporting a method's wins in a region of the quality-diversity trade-off that is dominated by other achievable settings is not specific to this paper; it is common practice in the sampling literature. We highlight it here only because the original paper's claim of a favorable trade-off rests on exactly such a region.

3 Extending `Min-p`'s NLP Benchmark Evaluations

We next turned to the original paper's NLP benchmark evaluations of several models on GSM8K with Chain-of-Thought (Cobbe et al., 2021) and GPQA (5-shot) (Rein et al., 2023), which concluded that "`min-p` sampling achieves superior performance across benchmarks and temperatures."

3.1 Thorough Hyperparameter Sweep on GSM8K Contradicts Claim of `Min-p`'s Superiority

To test whether `min-p` indeed achieves superior performance, we conducted an extensive analysis on GSM8K CoT, sweeping the following models, samplers, hyperparameters and sampling seeds:

- **9 Models:** Qwen 2.5 (Qwen et al., 2025) 0.5B, 1.5B, 3B and 7B; Mistral 7Bv0.1 (Jiang et al., 2023); Llama (Grattafiori et al., 2024) 3.1 8B and 3.2 3B; Gemma 2 (Team et al., 2024) 2B and 9B.

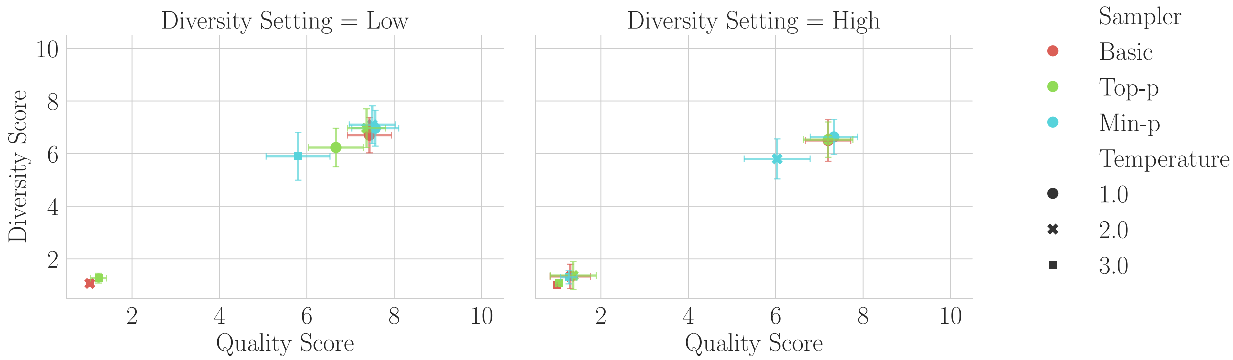

Figure 3: **New Human Evaluation Study Suggests `Min-p` Does Not Outperform Baselines in Quality, in Diversity or in a Pareto-Optimal Tradeoff Between Quality and Diversity.** Visualization of scores from Nguyen et al. (2024)'s second human experiment. `Min-p`'s performance advantage relative to `basic` and `top-p` sampling is observed in conditions (e.g., higher temperatures) where absolute quality and absolute diversity scores across all samplers are lower compared to other regimes (e.g., $T = 1$). For practitioners optimizing for maximal quality and maximal diversity, these results suggest that `min-p` offers no apparent advantage over `basic` or `top-p` sampling.

- **2 Model Stages:** Pre-trained ("Base") and Post-Trained ("Instruct").
- **4 Samplers:** `basic`, `top-p`, `top-k`, `min-p`.
- **31 Temperatures:** 0.0 ("greedy") to 3.0 in increments of 0.1.
- **6 Hyperparameters Per Sampler:** We chose 6 hyperparameters per sampler, except for `basic` which has no hyperparameter beyond temperature. The values were taken from the original paper's text, appendix tables and released evaluation logs, then extended to 6 values per sampler:
  - `basic`: No hyperparameters other than temperature.
  - `top-k`: $k \in \{10, 30, 50, 100, 150, 200\}$. The original paper tested $k \in \{10, 15, 20, 40, 50, 180\}$, which clusters four values between 10 and 50; we kept 10 and 50 and spread the remaining values more evenly up to 200.
  - `top-p`: $p \in \{0.99, 0.98, 0.95, 0.9, 0.8, 0.7\}$. The original paper's four values $\{0.7, 0.8, 0.9, 0.95\}$ were retained and 0.98 and 0.99 were added toward the no-truncation end.
  - `min-p`: $p \in \{0.01, 0.02, 0.05, 0.1, 0.2, 0.3\}$. The original paper's four values $\{0.05, 0.1, 0.2, 0.3\}$ were retained and 0.01 and 0.02 were added, since the original paper describes small p as the sensitive regime.

  Temperatures were densified from the original paper's $\{0.7, 1.0, 1.5, 2.0, 3.0\}$ to 0.1 spacing, and every sampler value was run at every temperature, whereas the original paper's released logs cover a ragged subset of the grid with one run per configuration.
- **3 Random Seeds for Sampling:** $\{0, 1, 2\}$

We evaluated GSM8K CoT under two prompt formats (for reasons explained below) and, in Sec. 3.2, GPQA. The GSM8K CoT sweeps required $\sim$ 6000 Nvidia A100-hours and the GPQA sweep $\sim$ 700 Nvidia A100-hours (summed run times, one GPU per run). GSM8K contains a subset of samples with ambiguous language or incorrect labels that have since been identified and cleaned (Vendrow et al., 2025) and that models may have been trained on GSM8K (Zhang et al., 2024), but we used GSM8K nonetheless for consistency with the original paper. We similarly used EleutherAI's LM Eval Harness (Gao et al., 2021; Biderman et al., 2024).

To evaluate how performant each sampler is, we first averaged over the three sampling seeds and then conducted two complementary analyses:

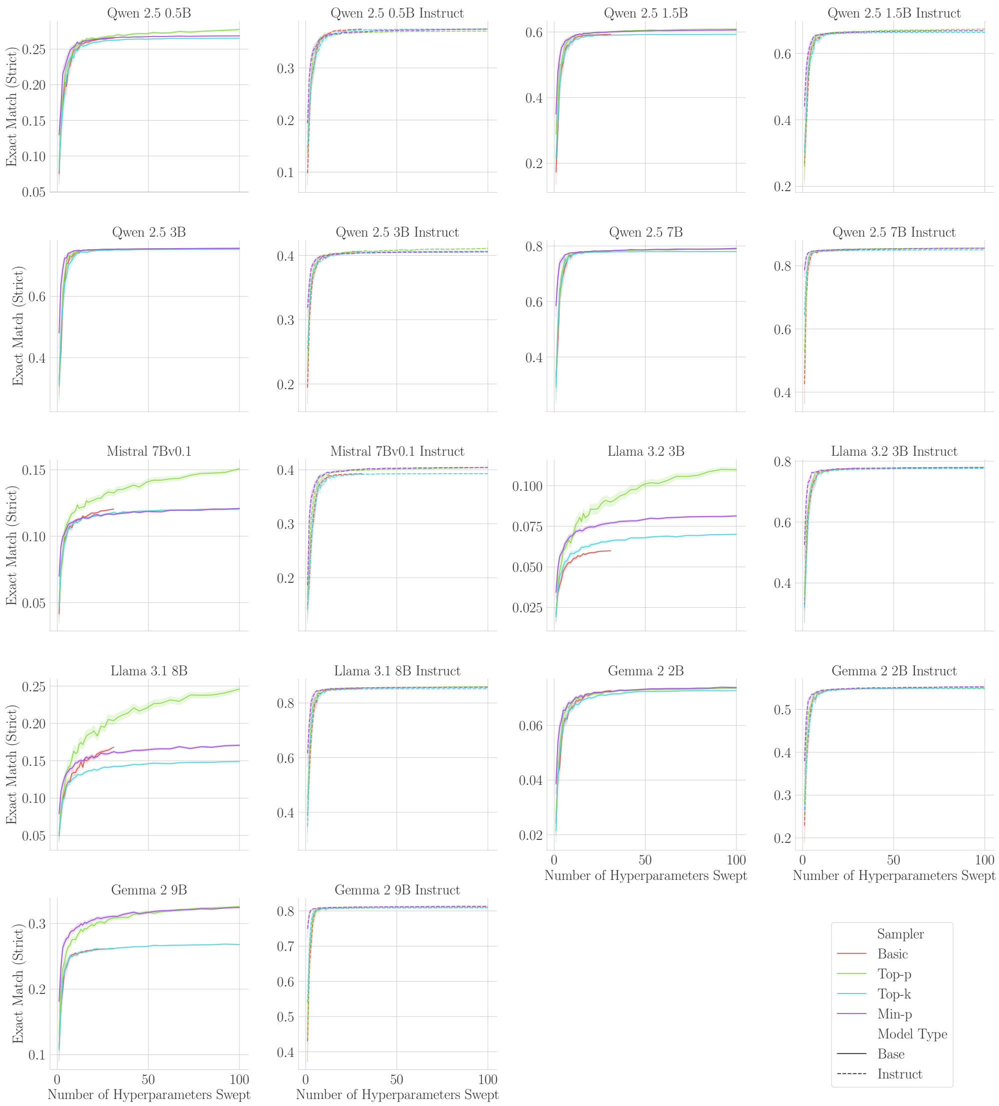

Figure 4: **`Min-P` Does Not Consistently Outperform Other Samplers on GSM8K When Controlling For Hyperparameter Volume.** In our first analysis, we measured how the maximum Exact Match (Strict) for each sampler improves as the number of hyperparameters increases. `Basic` sampling has only a temperature hyperparameter, and we therefore do not sweep it to the same degree.

1. For each sampler, we subsampled an equal number of hyperparameters ranging from $N = 1$ to $N = 100$ and computed the maximum Exact Match (Strict) score achieved by the sampled subset of size N. We repeated this process 150 times, averaging over the subsampled subsets' scores. This "Best-of-N" analysis (Nakano et al., 2021; Stiennon et al., 2020; Hughes et al., 2024; Schaeffer et al., 2025a) tells us the best possible performance each sampler will likely obtain as its hyperparameter space increases.

2. For $N = 1$ to $N = 100$, we subsampled N hyperparameters per sampler and computed the difference of the maximum Exact Match (Strict) score achieved by `min-p` minus the maximum score achieved by any other sampler. We repeated this process 150 times, averaging over the subsampled subsets. This tells us by how much `min-p` outperforms all other samplers, controlling for the size of hyperparameter space of each sampler.

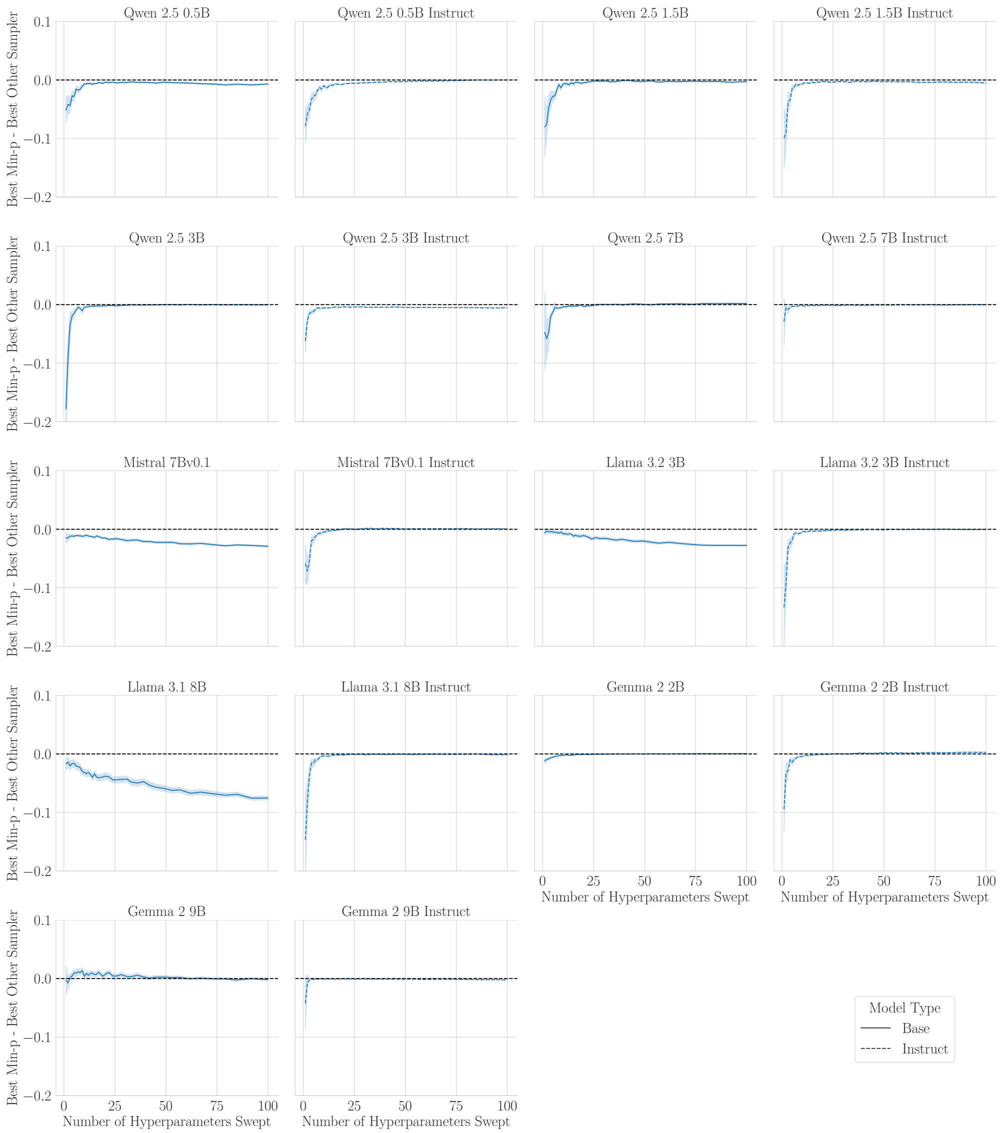

Figure 5: **`Min-P` Does Not Consistently Outperform Other Samplers on GSM8K When Controlling For Hyperparameter Volume.** In our second analysis, we measured how the difference between `min-p`'s highest score and the best non-`min-p` sampler's highest score changes as the number of swept hyperparameters increases. `Min-p` matches or underperforms other samplers.

Both analyses reached consistent results: `min-p` does not outperform other samplers when equalizing the volume of hyperparameter space. Fig. 4 and Fig. 5 respectively demonstrate that min-p is largely indistinguishable from other samplers.

After we showed these results to the authors, they informed us that we had run our experiments using the "Llama" formatting of GSM8K CoT prompts as we used the command from the authors' public Colab notebook; the authors clarified that "Llama" formatting should be used only for Llama models. We then reran our experiments using standard formatting of GSM8K CoT prompts. The results were nearly identical (Appendix B), with one small difference: `min-p` does produce higher scores for 2 of 12 language models. Again, we conclude **`min-p` does not outperform other samplers on either formatting of GSM8K CoT when controlling for hyperparameter volume**.

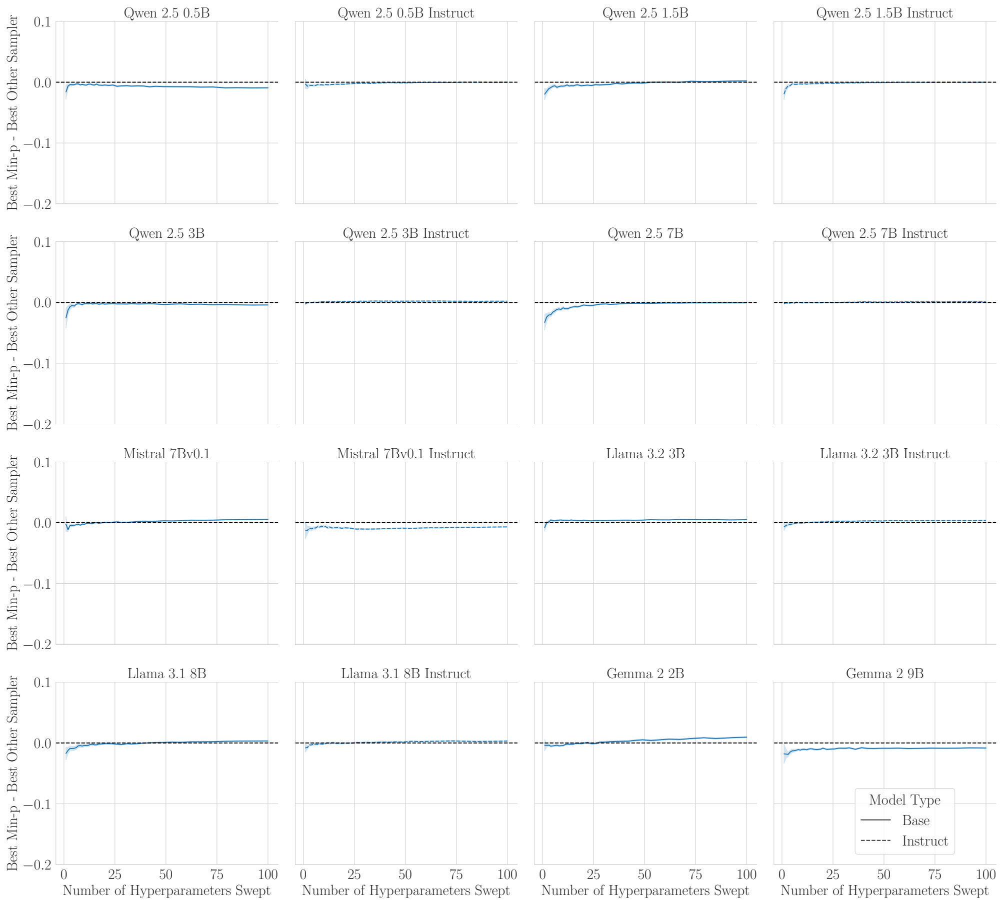

Figure 6: **`Min-P` Does Not Consistently Outperform Other Samplers on GPQA When Controlling For Hyperparameter Volume.** Difference between the best-of-N score of `min-p` and the best-of-N score of the best other sampler on GPQA (5-shot, flexible-extract exact match), using the same models, samplers, temperatures, sampler values and seeds as the GSM8K CoT sweep. The difference stays within roughly one percentage point at every N for all 16 models: `min-p` is slightly below the best other sampler on three models, slightly above on two, and indistinguishable on the rest. Gemma 2 2B Instruct and Gemma 2 9B Instruct are omitted because their GPQA runs produced no scores under the pinned `lm-eval` version. Best-of-N scores per sampler are shown in Appendix C.

3.2 Thorough Hyperparameter Sweep on GPQA Reaches the Same Conclusion

The original paper also evaluated GPQA (Rein et al., 2023), and its claim of superiority was stated "across benchmarks." We therefore repeated the sweep of Sec. 3.1 on GPQA (5-shot) with the identical grid of models, samplers, temperatures, sampler values and seeds, and applied the same two best-of-N analyses. Two Gemma 2 Instruct models produced no GPQA scores under the pinned `lm-eval` version and are omitted, leaving 16 models. The results (Fig. 6; Appendix C) match those on GSM8K CoT: once every sampler is given the same number of hyperparameter configurations, the best `min-p` configuration is within roughly one percentage point of the best other sampler on every model. **On both of the original paper's NLP benchmarks, `min-p` does not outperform other samplers when controlling for hyperparameter volume.**

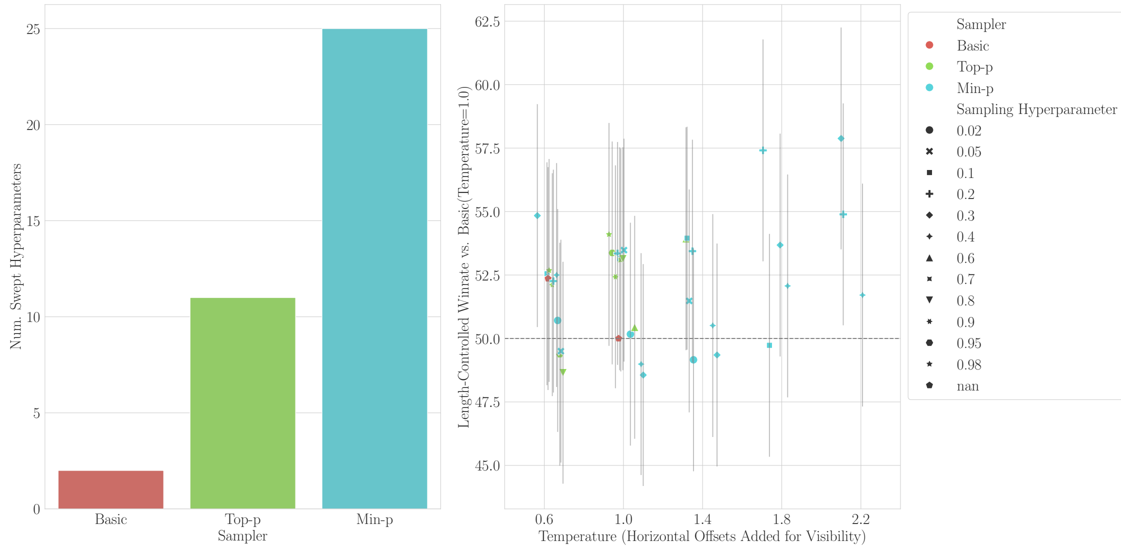

Figure 7: **Nguyen et al. (2024)'s LLM-As-A-Judge Evaluations Suggest `Min-p` Typically Matches Other Samplers Despite $2\times$ to $10\times$ More Hyperparameter Tuning.** Left: Nguyen et al. (2024) swept `min-p` with more than twice as many hyperparameters as `top-p` and more than ten times as many hyperparameters as `basic`. Right: Pairwise comparisons show `min-p` typically performs on-par with other samplers. Data were obtained from the first author's public GitHub repository.

4 Investigating `Min-p`'s LLM-As-A-Judge Evaluations

We then turned to the original paper's LLM-as-a-Judge evaluations (Zheng et al., 2023), specifically AlpacaEval creative writing evaluations (Dubois et al., 2023).

4.1 Under-Specified and Indirect Methodology Hinders Reproduction and Interpretation

In the Oct 2024 Arxiv manuscript and ICLR OpenReview manuscript, the methodology is under-specified in several ways: There is no mention which model(s) were sampled from, which model(s) served as the judge(s), or how hyperparameters were chosen or swept. Additionally, there is no description of uncertainty for the reported win rates, meaning readers are unable to decide whether win rates are statistically different from chance (50.00%).

Furthermore, the experimental design is indirect. AlpacaEval reports win rates between paired comparisons. Instead of directly comparing `min-p` against other samplers, the authors compared each sampler against a common fixed sampler: `basic`$(\tau = 1.0)$. Consequently, `min-p` was never tested head-to-head against `top-p`, or against `basic` at any other temperature, even though direct comparisons would have offered a clearer test of its superiority using the same number of comparisons.

Separately, subsequent work has shown that LLM-judge preferences are often not transitive (Xu et al., 2025): if sampler A beats sampler C and sampler B beats sampler C, it does not follow that A beats B. We note that Xu et al. (2025) was posted after the ICLR 2025 review cycle concluded, so the original authors could not have been expected to account for it; the point concerns what the reported comparisons can support today, not the original design choice. Under non-transitivity, comparing every sampler to `basic`$(\tau = 1.0)$ cannot in general be chained into conclusions about `min-p` relative to `top-p` or `basic` at other temperatures. **The under-specified methodology and the indirect experimental design together make drawing conclusions difficult.**

4.2 `Min-p` Received More Hyperparameter Tuning and Frequently Fails to Win

No scores or code to create scores were provided with the original paper's GitHub repository. While drafting this manuscript, we became aware of ongoing work to release code in a separate repository. Results from that code revealed two discoveries: First, `min-p` received $\sim 2\times$ more hyperparameter tuning than `top-p` sampling and $\sim 10\times$ more tuning than `basic` sampling (Fig. 7, left); a larger search alone would be expected to raise a sampler's best reported win rate. Second, the win-rates show that `min-p` frequently fails to outperform `top-p` and `basic` sampling, especially when accounting for confidence intervals; we visualized the new data with 95% confidence intervals (with horizontal offsets added for visibility) (Fig. 7, right).

4.3 Table 3(b) Reported The Higher of Two Scores For `Min-p` But the Lower of Two Scores For `Top-p`

As evidence for the LLM-As-A-Judge evaluation scores in the original paper's Table 3(b), the first author publicly shared a Telegram link that showed the higher of two scores was reported for `min-p` (the reported win rate of 52.01 corresponds to $p = 0.05$, but $p = 0.01$ yields a lower win rate of 50.14) but the lower of two score was reported for `top-p` (the reported win rate of 50.07 corresponds to $p = 0.9$, but $p = 0.98$ yields a higher win rate of 50.43).

5 Substantiating `Min-p`'s Community Adoption Claims

5.1 Claimed GitHub Repositories & Stars Were Unsubstantiated and Retracted

The Arxiv and peer-reviewed manuscripts of Nguyen et al. (2024) included specific claims about `min-p`'s adoption in the language modeling community:

> "**Community Adoption:** Min-p sampling has been rapidly adopted by the open-source community, with over 54,000 GitHub repositories using it, amassing a cumulative 1.1 million stars across these projects."

We attempted to verify these numbers through analysis of major GitHub language modeling repositories. Per our calculations, the combined GitHub stars of leading LM repositories (`transformers`, `ollama`, `llama.cpp`, `vLLM`, `Unsloth`, `mamba`, `SGLang`, `llama-cpp-python`) sum to 453k stars as of March 2025, less than half the 1.1M stars claimed by `min-p` alone. We could not substantiate either 49k GitHub repositories or 1.1M GitHub stars. When we inquired how these numbers were calculated, the authors publicly stated that GitHub was searched for "min-p", which yields many false positives. **The authors retracted both the 54k GitHub repository claim and the 1.1M GitHub stars claim from the ICLR 2025 Camera Ready manuscript.**

Although the numbers have been retracted, we document them here for two reasons. First, the figures were not supported in the manuscript, yet they circulated widely: readers may have encountered them when the paper was posted to arXiv or accepted at ICLR 2025, and they were cited as evidence of impact in the public ICLR 2025 review record (Appendix E). Second, as we detail below, the revised community adoption statement reports the usage of the frameworks that integrate `min-p` rather than the usage of `min-p` itself.

5.2 The Revised Community Adoption Statement Reports Framework Usage Rather Than `Min-p` Usage

The ICLR 2025 Camera Ready now has a different statement of community adoption:

> "[`Min-p`] is now integrated in widely used frameworks such as Hugging Face Transformers, vLLM, and SGLang, which collectively have accrued over 350,000 GitHub stars. This integration, coupled with extensive downstream usage (e.g., over 290,000 dependent repositories for Transformers alone), underscores the method's practical impact."

Being integrated into such frameworks is a genuine contribution. However, the stars and dependent repositories cited belong to the frameworks as a whole; they do not measure how many users of those frameworks enable `min-p`, so the statement does not quantify `min-p`'s adoption.

6 Discussion and Limitations

Scientific Conclusions This investigation led us to conclude that the four lines of evidence presented by Nguyen et al. (2024) – (1) human evaluations, (2) NLP benchmark evaluations, (3) LLM-as-a-Judge evaluations, (4) community adoption – do not support claims of `min-p`'s superiority. While `min-p` is useful for providing users another option to try, the original paper's data and our extensions of the original paper's data suggest that all samplers perform roughly the same once given the same amount of hyperparameter tuning; however, in our view, more research would be needed to assess the veracity of this conclusion. One of the twelve human evaluation comparisons survives Bonferroni correction: quality, `min-p` versus `basic`, at $\tau = 3.0$ (Table 1). At that temperature every sampler, including `min-p`, receives lower quality and diversity scores than at standard temperatures, so this is not an advantage a practitioner could use. We therefore do not read it as evidence that `min-p` improves quality or diversity.

Which Issues Are Specific to This Paper? Several of the practices we criticize are widespread rather than unique to Nguyen et al. (2024): reporting wins in a dominated region of a quality-diversity trade-off, testing many comparisons without correcting for multiplicity, and using indirect LLM-as-a-judge designs against a fixed reference. We flag them because the original paper's conclusions depend on them, and we hope the community will treat them as reasons to raise its standards rather than as faults of any one group. Other issues are specific to this paper: omitting collected `basic` sampler data from the human evaluation, reporting the higher of two scores for `min-p` but the lower of two for `top-p` in Table 3(b), and stating community adoption figures that could not be substantiated and were later retracted.

Key Limitation Our manuscript re-analyzes the evidence presented by the original paper (Nguyen et al., 2024) and additional evidence created using the original paper's code. *Conclusions here are based on that evidence.* We emphasize that new evidence might lead to different conclusions. Our own benchmark sweeps cover the two NLP benchmarks in the original paper, GSM8K CoT and GPQA, on models released through 2024 (up to 72B parameters for GSM8K CoT; Appendix D). GSM8K and GPQA reward accuracy rather than diversity, so they cannot reveal a benefit that `min-p` might provide where diversity is instrumentally useful, such as Minimum Bayes Risk decoding (Freitag et al., 2023), or in open-ended tasks beyond the creative writing evaluated by the human and LLM-as-a-judge studies. Newer models, additional benchmarks and such downstream uses remain open directions.

Relation to the ICLR 2025 Review Record Several of the issues documented here, in particular the retracted adoption figures and the unreported uncertainty in the LLM-as-a-judge win rates, bear on claims that were cited in the public ICLR 2025 reviews and meta-review of Nguyen et al. (2024). We summarize the relevant parts of that record in Appendix E.

References

David H Ackley, Geoffrey E Hinton, and Terrence J Sejnowski. A learning algorithm for boltzmann machines. *Cognitive science*, 9(1):147–169, 1985.

Sourya Basu, Govardana Sachitanandam Ramachandran, Nitish Shirish Keskar, and Lav R Varshney. Mirostat: A neural text decoding algorithm that directly controls perplexity. *arXiv preprint arXiv:2007.14966*, 2020.

Stella Biderman, Hailey Schoelkopf, Lintang Sutawika, Leo Gao, Jonathan Tow, Baber Abbasi, Alham Fikri Aji, Pawan Sasanka Ammanamanchi, Sidney Black, Jordan Clive, Anthony DiPofi, Julen Etxaniz, Benjamin Fattori, Jessica Zosa Forde, Charles Foster, Jeffrey Hsu, Mimansa Jaiswal, Wilson Y. Lee, Haonan Li, Charles Lovering, Niklas Muennighoff, Ellie Pavlick, Jason Phang, Aviya Skowron, Samson Tan, Xiangru Tang, Kevin A. Wang, Genta Indra Winata, François Yvon, and Andy Zou. Lessons from the trenches on reproducible evaluation of language models, 2024. URL https://arxiv.org/abs/2405.14782.

Wei-Lin Chiang, Lianmin Zheng, Ying Sheng, Anastasios Nikolas Angelopoulos, Tianle Li, Dacheng Li, Hao Zhang, Banghua Zhu, Michael Jordan, Joseph E. Gonzalez, and Ion Stoica. Chatbot arena: An open platform for evaluating llms by human preference, 2024. URL https://arxiv.org/abs/2403.04132.

Elizabeth Clark, Tal August, Sofia Serrano, Nikita Haduong, Suchin Gururangan, and Noah A Smith. All that's 'human'is not gold: Evaluating human evaluation of generated text. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, pp. 7282–7296, 2021.

Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, Christopher Hesse, and John Schulman. Training verifiers to solve math word problems, 2021. URL https://arxiv.org/abs/2110.14168.

Yann Dubois, Chen Xuechen Li, Rohan Taori, Tianyi Zhang, Ishaan Gulrajani, Jimmy Ba, Carlos Guestrin, Percy S Liang, and Tatsunori B Hashimoto. Alpacafarm: A simulation framework for methods that learn from human feedback. *Advances in Neural Information Processing Systems*, 36:30039–30069, 2023.

Angela Fan, Mike Lewis, and Yann Dauphin. Hierarchical neural story generation. *arXiv preprint arXiv:1805.04833*, 2018.

Markus Freitag, Behrooz Ghorbani, and Patrick Fernandes. Epsilon sampling rocks: Investigating sampling strategies for minimum bayes risk decoding for machine translation. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, 2023.

Leo Gao, Jonathan Tow, Stella Biderman, Sid Black, Anthony DiPofi, Charles Foster, Laurence Golding, Jeffrey Hsu, Kyle McDonell, Niklas Muennighoff, et al. A framework for few-shot language model evaluation. *Version v0. 0.1. Sept*, 10:8–9, 2021.

Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Alex Vaughan, Amy Yang, Angela Fan, Anirudh Goyal, Anthony Hartshorn, Aobo Yang, Archi Mitra, Archie Sravankumar, Artem Korenev, Arthur Hinsvark, Arun Rao, Aston Zhang, Aurelien Rodriguez, Austen Gregerson, Ava Spataru, Baptiste Roziere, Bethany Biron, Binh Tang, Bobbie Chern, Charlotte Caucheteux, Chaya Nayak, Chloe Bi, Chris Marra, Chris McConnell, Christian Keller, Christophe Touret, Chunyang Wu, Corinne Wong, Cristian Canton Ferrer, Cyrus Nikolaidis, Damien Allonsius, Daniel Song, Danielle Pintz, Danny Livshits, Danny Wyatt, David Esiobu, Dhruv Choudhary, Dhruv Mahajan, Diego Garcia-Olano, Diego Perino, Dieuwke Hupkes, Egor Lakomkin, Ehab AlBadawy, Elina Lobanova, Emily Dinan, Eric Michael Smith, Filip Radenovic, Francisco Guzmán, Frank Zhang, Gabriel Synnaeve, Gabrielle Lee, Georgia Lewis Anderson, Govind Thattai, Graeme Nail, Gregoire Mialon, Guan Pang, Guillem Cucurell, Hailey Nguyen, Hannah Korevaar, Hu Xu, Hugo Touvron, Iliyan Zarov, Imanol Arrieta Ibarra, Isabel Kloumann, Ishan Misra, Ivan Evtimov, Jack Zhang, Jade Copet, Jaewon Lee, Jan Geffert, Jana Vranes, Jason Park, Jay Mahadeokar, Jeet Shah, Jelmer van der Linde, Jennifer Billock, Jenny Hong, Jenya Lee, Jeremy Fu, Jianfeng Chi, Jianyu Huang, Jiawen Liu, Jie Wang, Jiecao Yu, Joanna Bitton, Joe Spisak, Jongsoo Park, Joseph Rocca, Joshua Johnstun, Joshua Saxe, Junteng Jia, Kalyan Vasuden Alwala, Karthik Prasad, Kartikeya Upasani, Kate Plawiak, Ke Li, Kenneth Heafield, Kevin Stone, Khalid El-Arini, Krithika Iyer, Kshitiz Malik, Kuenley Chiu, Kunal Bhalla, Kushal Lakhotia, Lauren Rantala-Yeary, Laurens van der Maaten, Lawrence Chen, Liang Tan, Liz Jenkins, Louis Martin, Lovish Madaan, Lubo Malo, Lukas Blecher, Lukas Landzaat, Luke de Oliveira, Madeline Muzzi, Mahesh Pasupuleti, Mannat Singh, Manohar Paluri, Marcin Kardas, Maria Tsimpoukelli, Mathew Oldham, Mathieu Rita, Maya Pavlova, Melanie Kambadur, Mike Lewis, Min Si, Mitesh Kumar Singh, Mona Hassan, Naman Goyal, Narjes Torabi, Nikolay Bashlykov, Nikolay Bogoychev, Niladri Chatterji, Ning Zhang, Olivier Duchenne, Onur Çelebi, Patrick Alrassy, Pengchuan Zhang, Pengwei Li, Petar Vasic, Peter Weng, Prajjwal Bhargava, Pratik Dubal, Praveen Krishnan, Punit Singh Koura, Puxin Xu, Qing He, Qingxiao Dong, Ragavan Srinivasan, Raj Ganapathy, Ramon Calderer, Ricardo Silveira Cabral, Robert Stojnic, Roberta Raileanu, Rohan Maheswari, Rohit Girdhar, Rohit Patel, Romain Sauvestre, Ronnie Polidoro, Roshan Sumbaly, Ross Taylor, Ruan Silva, Rui Hou, Rui Wang, Saghar Hosseini, Sahana Chennabasappa, Sanjay Singh, Sean Bell, Seohyun Sonia Kim, Sergey Edunov, Shaoliang Nie, Sharan Narang, Sharath Raparthy, Sheng Shen, Shengye Wan, Shruti Bhosale, Shun Zhang, Simon Vandenhende, Soumya Batra, Spencer Whitman, Sten Sootla, Stephane Collot, Suchin Gururangan, Sydney Borodinsky, Tamar Herman, Tara Fowler, Tarek Sheasha, Thomas Georgiou, Thomas Scialom, Tobias Speckbacher, Todor Mihaylov, Tong Xiao, Ujjwal Karn, Vedanuj Goswami, Vibhor Gupta, Vignesh Ramanathan, Viktor Kerkez, Vincent Gonguet, Virginie Do, Vish Vogeti, Vítor Albiero, Vladan Petrovic, Weiwei Chu, Wenhan Xiong, Wenyin Fu, Whitney Meers, Xavier Martinet, Xiaodong Wang, Xiaofang Wang, Xiaoqing Ellen Tan, Xide Xia, Xinfeng Xie, Xuchao Jia, Xuewei Wang, Yaelle Goldschlag, Yashesh Gaur, Yasmine Babaei, Yi Wen, Yiwen Song, Yuchen Zhang, Yue Li, Yuning Mao, Zacharie Delpierre Coudert, Zheng Yan, Zhengxing Chen, Zoe Papakipos, Aaditya Singh, Aayushi Srivastava, Abha Jain, Adam Kelsey, Adam Shajnfeld, Adithya Gangidi, Adolfo Victoria, Ahuva Goldstand, Ajay Menon, Ajay Sharma, Alex Boesenberg, Alexei Baevski, Allie Feinstein, Amanda Kallet, Amit Sangani, Amos Teo, Anam Yunus, Andrei Lupu, Andres Alvarado, Andrew Caples, Andrew Gu, Andrew Ho, Andrew Poulton, Andrew Ryan, Ankit Ramchandani, Annie Dong, Annie Franco, Anuj Goyal, Aparajita Saraf, Arkabandhu Chowdhury, Ashley Gabriel, Ashwin Bharambe, Assaf Eisenman, Azadeh Yazdan, Beau James, Ben Maurer, Benjamin Leonhardi, Bernie Huang, Beth Loyd, Beto De Paola, Bhargavi Paranjape, Bing Liu, Bo Wu, Boyu Ni, Braden Hancock, Bram Wasti, Brandon Spence, Brani Stojkovic, Brian Gamido, Britt Montalvo, Carl Parker, Carly Burton, Catalina Mejia, Ce Liu, Changhan Wang, Changkyu Kim, Chao Zhou, Chester Hu, Ching-Hsiang Chu, Chris Cai, Chris Tindal, Christoph Feichtenhofer, Cynthia Gao, Damon Civin, Dana Beaty, Daniel Kreymer, Daniel Li, David Adkins, David Xu, Davide Testuggine, Delia David, Devi Parikh, Diana Liskovich, Didem Foss, Dingkang Wang, Duc Le, Dustin Holland, Edward Dowling, Eissa Jamil, Elaine Montgomery, Eleonora Presani, Emily Hahn, Emily Wood, Eric-Tuan Le, Erik Brinkman, Esteban Arcaute, Evan Dunbar, Evan Smothers, Fei Sun, Felix Kreuk, Feng Tian, Filippos Kokkinos, Firat Ozgenel, Francesco Caggioni, Frank Kanayet, Frank Seide, Gabriela Medina Florez, Gabriella Schwarz, Gada Badeer, Georgia Swee, Gil Halpern, Grant Herman, Grigory Sizov, Guangyi, Zhang, Guna Lakshminarayanan, Hakan Inan, Hamid Shojanazeri, Han Zou, Hannah Wang, Hanwen Zha, Haroun Habeeb, Harrison Rudolph, Helen Suk, Henry Aspegren, Hunter Goldman, Hongyuan Zhan, Ibrahim Damlaj, Igor Molybog, Igor Tufanov, Ilias Leontiadis, Irina-Elena Veliche, Itai Gat, Jake Weissman, James Geboski, James Kohli, Janice Lam, Japhet Asher, Jean-Baptiste Gaya, Jeff Marcus, Jeff Tang, Jennifer Chan, Jenny Zhen, Jeremy Reizenstein, Jeremy Teboul, Jessica Zhong, Jian Jin, Jingyi Yang, Joe Cummings, Jon Carvill, Jon Shepard, Jonathan McPhie, Jonathan Torres, Josh Ginsburg, Junjie Wang, Kai Wu, Kam Hou U, Karan Saxena, Kartikay Khandelwal, Katayoun Zand, Kathy Matosich, Kaushik Veeraraghavan, Kelly Michelena, Keqian Li, Kiran Jagadeesh, Kun Huang, Kunal Chawla, Kyle Huang, Lailin Chen, Lakshya Garg, Lavender A, Leandro Silva, Lee Bell, Lei Zhang, Liangpeng Guo, Licheng Yu, Liron Moshkovich, Luca Wehrstedt, Madian Khabsa, Manav Avalani, Manish Bhatt, Martynas Mankus, Matan Hasson, Matthew Lennie, Matthias Reso, Maxim Groshev, Maxim Naumov, Maya Lathi, Meghan Keneally, Miao Liu, Michael L. Seltzer, Michal Valko, Michelle Restrepo, Mihir Patel, Mik Vyatskov, Mikayel Samvelyan, Mike Clark, Mike Macey, Mike Wang, Miquel Jubert Hermoso, Mo Metanat, Mohammad Rastegari, Munish Bansal, Nandhini Santhanam, Natascha Parks, Natasha White, Navyata Bawa, Nayan Singhal, Nick Egebo, Nicolas Usunier, Nikhil Mehta, Nikolay Pavlovich Laptev, Ning Dong, Norman Cheng, Oleg Chernoguz, Olivia Hart, Omkar Salpekar, Ozlem Kalinli, Parkin Kent, Parth Parekh, Paul Saab, Pavan Balaji, Pedro Rittner, Philip Bontrager, Pierre Roux, Piotr Dollar, Polina Zvyagina, Prashant Ratanchandani, Pritish Yuvraj, Qian

Liang, Rachad Alao, Rachel Rodriguez, Rafi Ayub, Raghotham Murthy, Raghu Nayani, Rahul Mitra, Rangaprabhu Parthasarathy, Raymond Li, Rebekkah Hogan, Robin Battey, Rocky Wang, Russ Howes, Ruty Rinott, Sachin Mehta, Sachin Siby, Sai Jayesh Bondu, Samyak Datta, Sara Chugh, Sara Hunt, Sargun Dhillon, Sasha Sidorov, Satadru Pan, Saurabh Mahajan, Saurabh Verma, Seiji Yamamoto, Sharadh Ramaswamy, Shaun Lindsay, Shaun Lindsay, Sheng Feng, Shenghao Lin, Shengxin Cindy Zha, Shishir Patil, Shiva Shankar, Shuqiang Zhang, Shuqiang Zhang, Sinong Wang, Sneha Agarwal, Soji Sajuyigbe, Soumith Chintala, Stephanie Max, Stephen Chen, Steve Kehoe, Steve Satterfield, Sudarshan Govindaprasad, Sumit Gupta, Summer Deng, Sungmin Cho, Sunny Virk, Suraj Subramanian, Sy Choudhury, Sydney Goldman, Tal Remez, Tamar Glaser, Tamara Best, Thilo Koehler, Thomas Robinson, Tianhe Li, Tianjun Zhang, Tim Matthews, Timothy Chou, Tzook Shaked, Varun Vontimitta, Victoria Ajayi, Victoria Montanez, Vijai Mohan, Vinay Satish Kumar, Vishal Mangla, Vlad Ionescu, Vlad Poenaru, Vlad Tiberiu Mihailescu, Vladimir Ivanov, Wei Li, Wenchen Wang, Wenwen Jiang, Wes Bouaziz, Will Constable, Xiaocheng Tang, Xiaojian Wu, Xiaolan Wang, Xilun Wu, Xinbo Gao, Yaniv Kleinman, Yanjun Chen, Ye Hu, Ye Jia, Ye Qi, Yenda Li, Yilin Zhang, Ying Zhang, Yossi Adi, Youngjin Nam, Yu, Wang, Yu Zhao, Yuchen Hao, Yundi Qian, Yunlu Li, Yuzi He, Zach Rait, Zachary DeVito, Zef Rosnbrick, Zhaoduo Wen, Zhenyu Yang, Zhiwei Zhao, and Zhiyu Ma. The llama 3 herd of models, 2024. URL `https://arxiv.org/abs/2407.21783`.

John Hewitt, Christopher D Manning, and Percy Liang. Truncation sampling as language model desmoothing. In *Findings of the Association for Computational Linguistics: EMNLP 2022*, pp. 3414–3427, 2022.

Ari Holtzman, Jan Buys, Li Du, Maxwell Forbes, and Yejin Choi. The curious case of neural text degeneration. In *International Conference on Learning Representations*, 2020.

David M Howcroft, Anya Belz, Miruna Clinciu, Dimitra Gkatzia, Sadid A Hasan, Saad Mahamood, Simon Mille, Emiel Van Miltenburg, Sashank Santhanam, and Verena Rieser. Twenty years of confusion in human evaluation: Nlg needs evaluation sheets and standardised definitions. In *13th International Conference on Natural Language Generation 2020*, pp. 169–182. Association for Computational Linguistics, 2020.

John Hughes, Sara Price, Aengus Lynch, Rylan Schaeffer, Fazl Barez, Sanmi Koyejo, Henry Sleight, Erik Jones, Ethan Perez, and Mrinank Sharma. Best-of-n jailbreaking, 2024. URL `https://arxiv.org/abs/2412.03556`.

Albert Q. Jiang, Alexandre Sablayrolles, Arthur Mensch, Chris Bamford, Devendra Singh Chaplot, Diego de las Casas, Florian Bressand, Gianna Lengyel, Guillaume Lample, Lucile Saulnier, Lélio Renard Lavaud, Marie-Anne Lachaux, Pierre Stock, Teven Le Scao, Thibaut Lavril, Thomas Wang, Timothée Lacroix, and William El Sayed. Mistral 7b, 2023. URL `https://arxiv.org/abs/2310.06825`.

Liwei Jiang, Yuanjun Chai, Margaret Li, Mickel Liu, Raymond Fok, Nouha Dziri, Yulia Tsvetkov, Maarten Sap, Alon Albalak, and Yejin Choi. Artificial hivemind: The open-ended homogeneity of language models (and beyond). In *Advances in Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track*, 2025. arXiv:2510.22954.

Daniel Khashabi, Gabriel Stanovsky, Jonathan Bragg, Nicholas Lourie, Jungo Kasai, Yejin Choi, Noah A. Smith, and Daniel S. Weld. Genie: Toward reproducible and standardized human evaluation for text generation, 2022. URL `https://arxiv.org/abs/2101.06561`.

Percy Liang, Rishi Bommasani, Tony Lee, Dimitris Tsipras, Dilara Soylu, Michihiro Yasunaga, Yian Zhang, Deepak Narayanan, Yuhuai Wu, Ananya Kumar, et al. Holistic evaluation of language models. *arXiv preprint arXiv:2211.09110*, 2022.

Reiichiro Nakano, Jacob Hilton, Suchir Balaji, Jeff Wu, Long Ouyang, Christina Kim, Christopher Hesse, Shantanu Jain, Vineet Kosaraju, William Saunders, et al. Webgpt: Browser-assisted question-answering with human feedback. *arXiv preprint arXiv:2112.09332*, 2021.

Minh Nguyen, Andrew Baker, Clement Neo, Allen Roush, Andreas Kirsch, and Ravid Shwartz-Ziv. Turning up the heat: Min-p sampling for creative and coherent llm outputs. *arXiv preprint arXiv:2407.01082*, 2024.

Qwen, :, An Yang, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chengyuan Li, Dayiheng Liu, Fei Huang, Haoran Wei, Huan Lin, Jian Yang, Jianhong Tu, Jianwei Zhang, Jianxin Yang, Jiaxi Yang, Jingren Zhou, Junyang Lin, Kai Dang, Keming Lu, Keqin Bao, Kexin Yang, Le Yu, Mei Li, Mingfeng Xue, Pei Zhang, Qin Zhu, Rui Men, Runji Lin, Tianhao Li, Tianyi Tang, Tingyu Xia, Xingzhang Ren, Xuancheng Ren, Yang Fan, Yang Su, Yichang Zhang, Yu Wan, Yuqiong Liu, Zeyu Cui, Zhenru Zhang, and Zihan Qiu. Qwen2.5 technical report, 2025. URL `https://arxiv.org/abs/2412.15115`.

David Rein, Betty Li Hou, Asa Cooper Stickland, Jackson Petty, Richard Yuanzhe Pang, Julien Dirani, Julian Michael, and Samuel R. Bowman. Gpqa: A graduate-level google-proof q&a benchmark, 2023. URL `https://arxiv.org/abs/2311.12022`.

Stephen Roller, Y-Lan Boureau, Jason Weston, Antoine Bordes, Emily Dinan, Angela Fan, David Gunning, Da Ju, Margaret Li, Spencer Poff, Pratik Ringshia, Kurt Shuster, Eric Michael Smith, Arthur Szlam, Jack Urbanek, and Mary Williamson. Open-domain conversational agents: Current progress, open problems, and future directions, 2020. URL `https://arxiv.org/abs/2006.12442`.

Rylan Schaeffer, Joshua Kazdan, John Hughes, Jordan Juravsky, Sara Price, Aengus Lynch, Erik Jones, Robert Kirk, Azalia Mirhoseini, and Sanmi Koyejo. How do large language monkeys get their power (laws)?, 2025a. URL `https://arxiv.org/abs/2502.17578`.

Rylan Schaeffer, Punit Singh Koura, Binh Tang, Ranjan Subramanian, Aaditya K Singh, Todor Mihaylov, Prajjwal Bhargava, Lovish Madaan, Niladri S. Chatterji, Vedanuj Goswami, Sergey Edunov, Dieuwke Hupkes, Sanmi Koyejo, and Sharan Narang. Correlating and predicting human evaluations of language models from natural language processing benchmarks, 2025b. URL `https://arxiv.org/abs/2502.18339`.

Nisan Stiennon, Long Ouyang, Jeffrey Wu, Daniel Ziegler, Ryan Lowe, Chelsea Voss, Alec Radford, Dario Amodei, and Paul F Christiano. Learning to summarize with human feedback. *Advances in neural information processing systems*, 33:3008–3021, 2020.

Runyan Tan, Shuang Wu, and Phillip Howard. p-less sampling: A robust hyperparameter-free approach for llm decoding. In *International Conference on Learning Representations (ICLR)*, 2026. arXiv:2509.23234.

Gemma Team, Morgane Riviere, Shreya Pathak, Pier Giuseppe Sessa, Cassidy Hardin, Surya Bhupatiraju, Léonard Hussenot, Thomas Mesnard, Bobak Shahriari, Alexandre Ramé, Johan Ferret, Peter Liu, Pouya Tafti, Abe Friesen, Michelle Casbon, Sabela Ramos, Ravin Kumar, Charline Le Lan, Sammy Jerome, Anton Tsitsulin, Nino Vieillard, Piotr Stanczyk, Sertan Girgin, Nikola Momchev, Matt Hoffman, Shantanu Thakoor, Jean-Bastien Grill, Behnam Neyshabur, Olivier Bachem, Alanna Walton, Aliaksei Severyn, Alicia Parrish, Aliya Ahmad, Allen Hutchison, Alvin Abdagic, Amanda Carl, Amy Shen, Andy Brock, Andy Coenen, Anthony Laforge, Antonia Paterson, Ben Bastian, Bilal Piot, Bo Wu, Brandon Royal, Charlie Chen, Chintu Kumar, Chris Perry, Chris Welty, Christopher A. Choquette-Choo, Danila Sinopalnikov, David Weinberger, Dimple Vijaykumar, Dominika Rogozińska, Dustin Herbison, Elisa Bandy, Emma Wang, Eric Noland, Erica Moreira, Evan Senter, Evgenii Eltyshev, Francesco Visin, Gabriel Rasskin, Gary Wei, Glenn Cameron, Gus Martins, Hadi Hashemi, Hanna Klimczak-Plucińska, Harleen Batra, Harsh Dhand, Ivan Nardini, Jacinda Mein, Jack Zhou, James Svensson, Jeff Stanway, Jetha Chan, Jin Peng Zhou, Joana Carrasqueira, Joana Iljazi, Jocelyn Becker, Joe Fernandez, Joost van Amersfoort, Josh Gordon, Josh Lipschultz, Josh Newlan, Ju yeong Ji, Kareem Mohamed, Kartikeya Badola, Kat Black, Katie Millican, Keelin McDonell, Kelvin Nguyen, Kiranbir Sodhia, Kish Greene, Lars Lowe Sjoesund, Lauren Usui, Laurent Sifre, Lena Heuermann, Leticia Lago, Lilly McNealus, Livio Baldini Soares, Logan Kilpatrick, Lucas Dixon, Luciano Martins, Machel Reid, Manvinder Singh, Mark Iverson, Martin Görner, Mat Velloso, Mateo Wirth, Matt Davidow, Matt Miller, Matthew Rahtz, Matthew Watson, Meg Risdal, Mehran Kazemi, Michael Moynihan, Ming Zhang, Minsuk Kahng, Minwoo Park, Mofi Rahman, Mohit Khatwani, Natalie Dao, Nenshad Bardoliwalla, Nesh Devanathan, Neta Dumai, Nilay Chauhan, Oscar Wahltinez, Pankil Botarda, Parker Barnes, Paul Barham, Paul Michel, Pengchong Jin, Petko Georgiev, Phil Culliton, Pradeep Kuppala, Ramona Comanescu, Ramona Merhej, Reena Jana, Reza Ardeshir Rokni, Rishabh Agarwal, Ryan Mullins, Samaneh Saadat, Sara Mc Carthy, Sarah Cogan, Sarah Perrin, Sébastien M. R. Arnold, Sebastian Krause, Shengyang Dai, Shruti Garg, Shruti Sheth, Sue Ronstrom, Susan Chan, Timothy Jordan, Ting Yu, Tom Eccles, Tom Hennigan, Tomas Kocisky, Tulsee Doshi, Vihan Jain, Vikas Yadav, Vilobh

Meshram, Vishal Dharmadhikari, Warren Barkley, Wei Wei, Wenming Ye, Woohyun Han, Woosuk Kwon, Xiang Xu, Zhe Shen, Zhitao Gong, Zichuan Wei, Victor Cotruta, Phoebe Kirk, Anand Rao, Minh Giang, Ludovic Peran, Tris Warkentin, Eli Collins, Joelle Barral, Zoubin Ghahramani, Raia Hadsell, D. Sculley, Jeanine Banks, Anca Dragan, Slav Petrov, Oriol Vinyals, Jeff Dean, Demis Hassabis, Koray Kavukcuoglu, Clement Farabet, Elena Buchatskaya, Sebastian Borgeaud, Noah Fiedel, Armand Joulin, Kathleen Kenealy, Robert Dadashi, and Alek Andreev. Gemma 2: Improving open language models at a practical size, 2024. URL `https://arxiv.org/abs/2408.00118`.

Chris Van Der Lee, Albert Gatt, Emiel Van Miltenburg, Sander Wubben, and Emiel Krahmer. Best practices for the human evaluation of automatically generated text. In *Proceedings of the 12th International Conference on Natural Language Generation*, pp. 355–368, 2019.

Joshua Vendrow, Edward Vendrow, Sara Beery, and Aleksander Madry. Do large language model benchmarks test reliability? *arXiv preprint arXiv:2502.03461*, 2025.

Yi Xu, Laura Ruis, Tim Rocktäschel, and Robert Kirk. Investigating non-transitivity in llm-as-a-judge, 2025. URL `https://arxiv.org/abs/2502.14074`.

Hugh Zhang, Jeff Da, Dean Lee, Vaughn Robinson, Catherine Wu, William Song, Tiffany Zhao, Pranav Raja, Charlotte Zhuang, Dylan Slack, et al. A careful examination of large language model performance on grade school arithmetic. *Advances in Neural Information Processing Systems*, 37:46819–46836, 2024.

Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric Xing, et al. Judging llm-as-a-judge with mt-bench and chatbot arena. *Advances in Neural Information Processing Systems*, 36:46595–46623, 2023.

A Examples of Human Qualitative Responses Favoring `Basic` Sampling Over `Min-P` Sampling

In Section 2.3, we described how qualitative responses from many human participants in the original paper's study favored `basic` sampling. Direct quotes from human evaluators favoring `basic` sampling are provided below. In the study, `basic` sampling was called "Model A"; for clarity, we substituted the pseudonyms for the actual sampling methods):

- "[`basic` sampling] on Temp 3.0 - High Diversity setting. The stories where [sic] more interenting [sic], felt more different compared to the others, which felt like the same ideia [sic] just in a different format."
- "I felt like [`basic` sampling] was most diverse and most interesting with it's [sic] descriptions of the characters and the setting. It appealed to me most and seemed to have less 'broken' sentences that didn't make sense. Descriptions were painterly [sic] and elaborate."
- "[`basic` sampling] was more engaging, it aroused my curiosity."
- "[`basic` sampling] provided more depth and easy to read for me and there was more diversity."
- "[`basic` sampling], they presented creative storytelling"
- "[`basic` sampling]. From the very beginning the verbiage and descriptions were very creative and vivid. And each story was unique"
- "I believe that [`basic` sampling] has provided stories with more differentiation overall than the other two models. From the point of view of creativity, all three models are more or less equivalent as they almost always talk about stories set in extraterrestrial worlds both from a physical and mental (dreams) point of view"
- "[`Basic` sampling]: Sample 2: Temperature Setting F (Temp 3.0 - High Diversity). The story was captivating, it took inside the mystical land and walked you right besides all the characters, you can even draw the characters from just th descriptions provided by the prompt. you Could even smell them, smell the setting and be at one with the setting."
- "I personally preferred [`basic` sampling] on the setting of creative, descriptive storytelling. I enjoyed how the writing was creative, showing imagination and a strong use of language. The stories were quite evocative, with intriguing settings and characters that helped to draw the reader in. I also appreciated the diversity of themes that were explored, from night weavers to dream manipulation and mysterious libraries, which kept the stories engaging and interesting."
- "Temporature setting C on [`basic` sampling] was the best. The story was fascinating and very engaging. I wanted to read more."
- "I prefered the first [`basic` sampling]. Tho [`basic` sampling] and C seem to be very head to head. But something about [`basic` sampling] seemed different in quality about it to me."

More quotes are in the original paper's data. We urge readers to draw their own conclusions.

B GSM8K Chain-of-Thought Scores with "Standard" Formatting

At the request of Nguyen et al. (2024), we reran our GSM8K Chain-of-Thought sweeps using "standard" formatting instead of "Llama" formatting. **Both analyses reached consistent results: `min-p` does not consistently outperform other samplers when controlling the volume of hyperparameter space.**

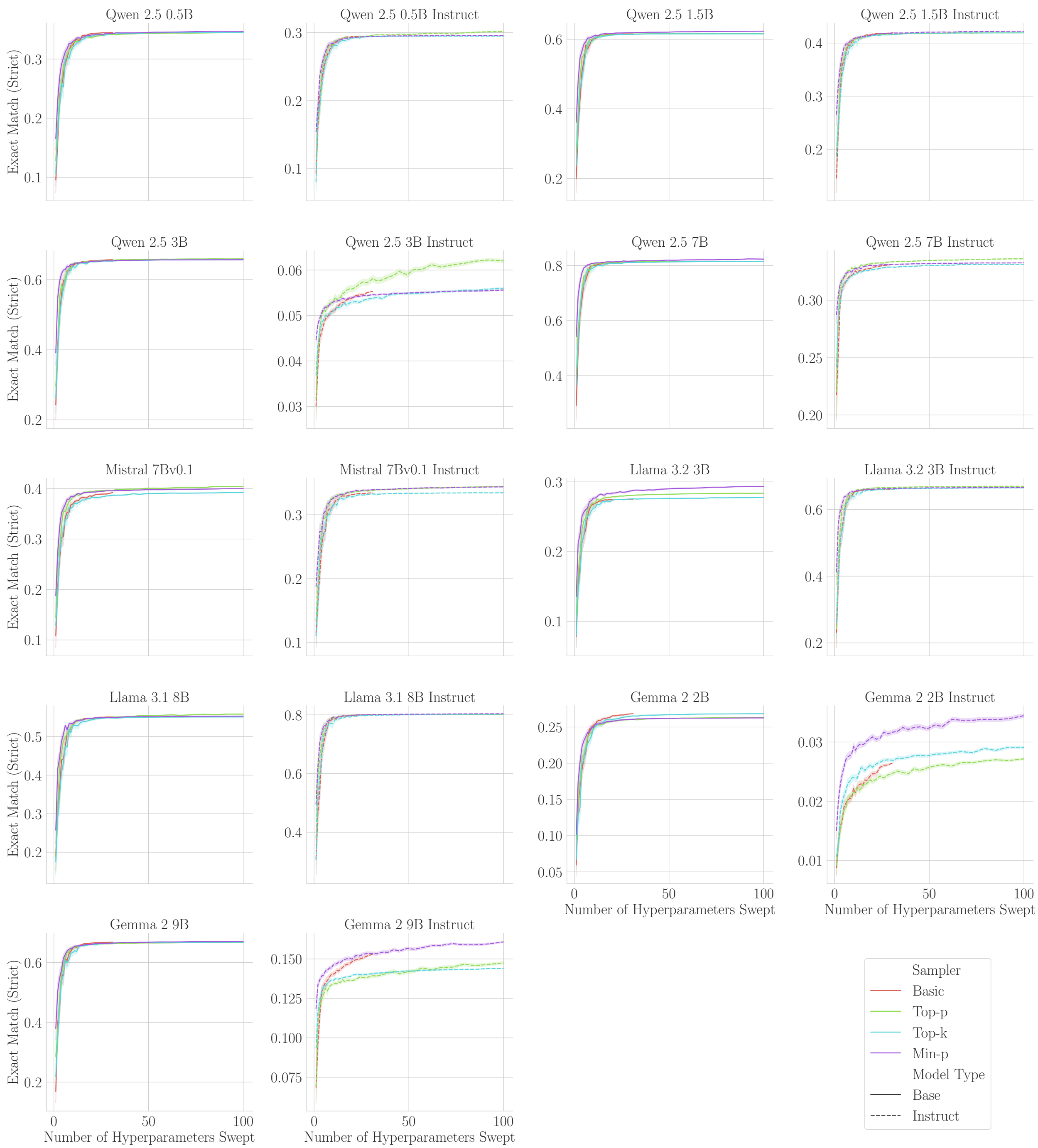

Figure 8: **`Min-P` Does Not Consistently Outperform Other Samplers on GSM8K When Controlling For Hyperparameter Volume.** We reran our GSM8K sweep using "standard" formatting rather than "Llama" formatting and observed qualitatively similar data.

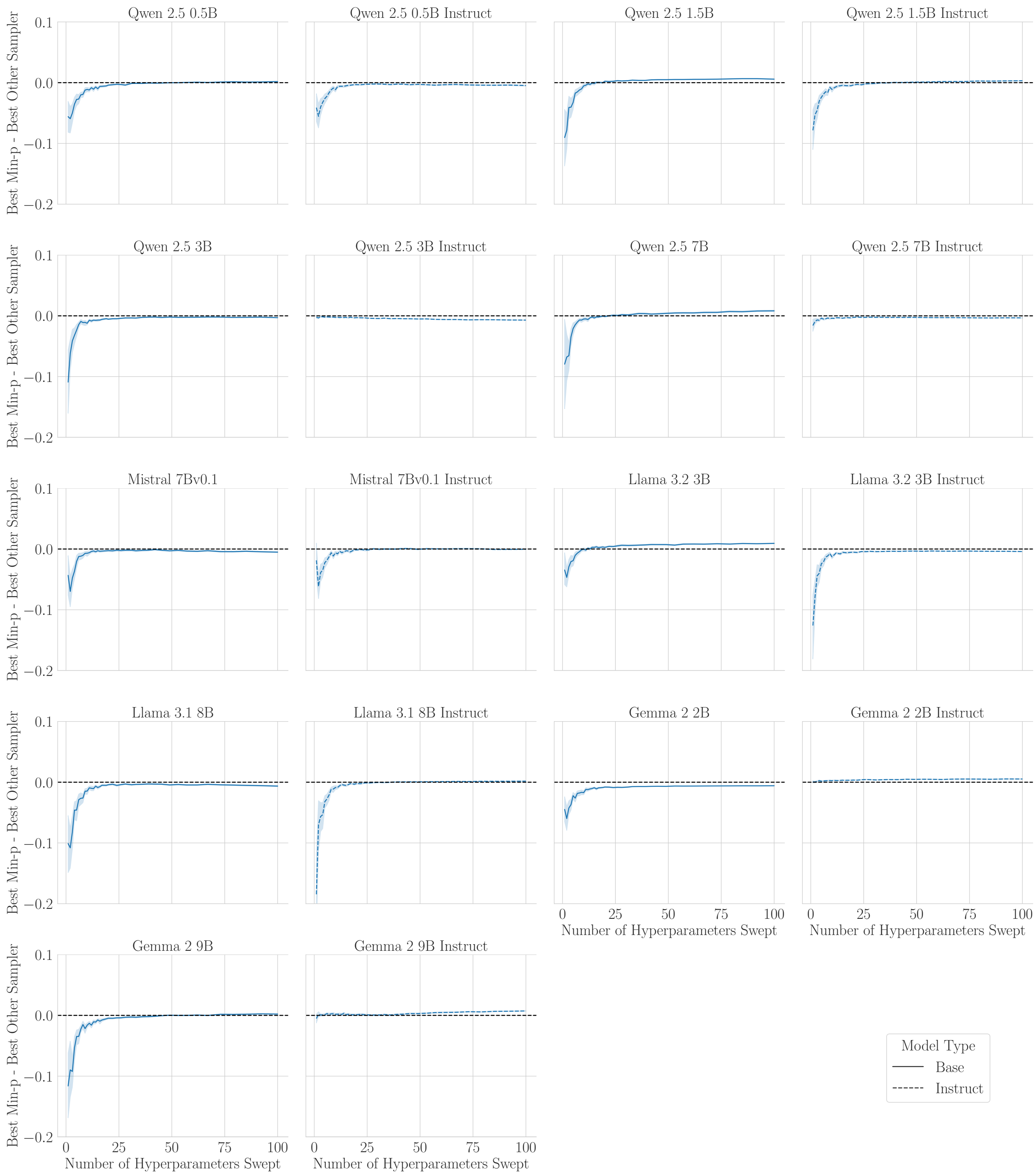

Figure 9: **`Min-P` Does Not Consistently Outperform Other Samplers on GSM8K When Controlling For Hyperparameter Volume.** We reran our GSM8K sweep using "standard" formatting rather than "Llama" formatting and observed qualitatively similar data.

C GPQA Best-of-N Scores

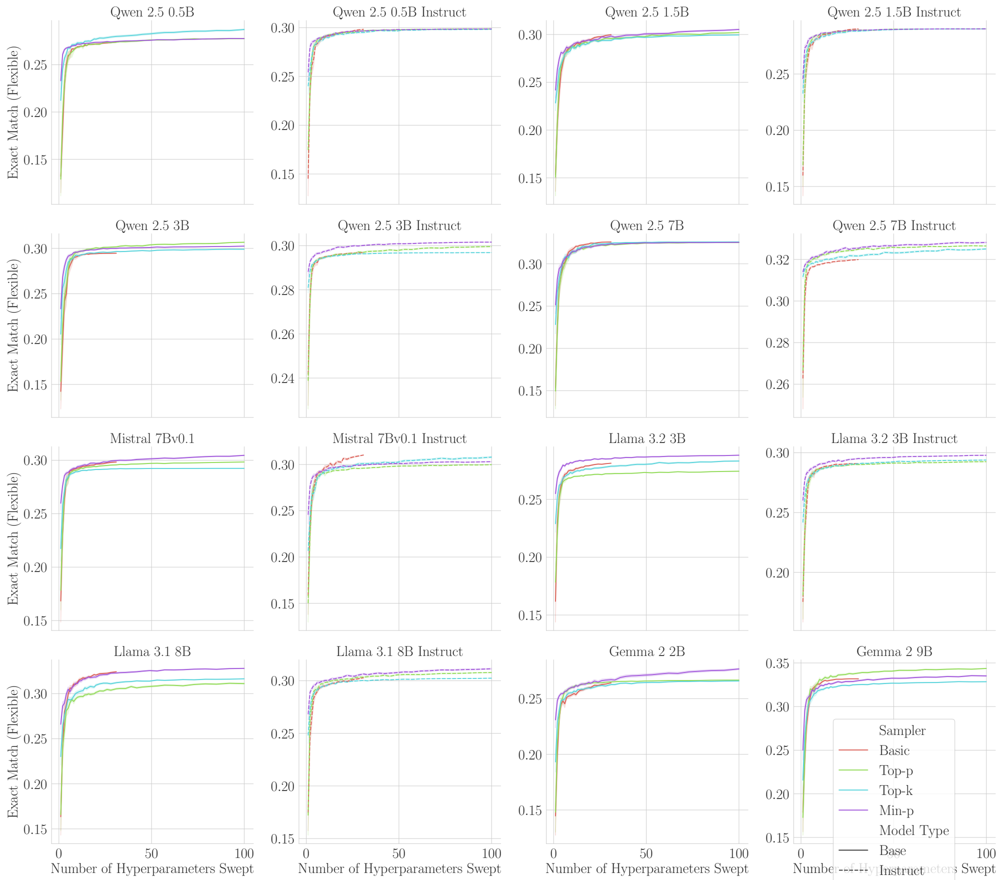

Figure 10: **Best-of-N GPQA Scores Per Sampler.** Companion to Fig. 6: the best-of-N flexible-extract exact match of each sampler on GPQA (5-shot) for the 16 models with complete sweeps. All four samplers reach similar scores once given the same number of hyperparameter configurations.

D GSM8K Chain-of-Thought Scores on Larger Models

To partially address whether our conclusions hold for larger models, we ran the GSM8K CoT sweep of Sec. 3.1 (standard formatting) on ten additional models: Qwen 2.5 14B, 32B and 72B, Gemma 2 27B, and Llama 3.1 70B, each in base and instruct variants. Fig. 11 shows the difference between the best-of-N score of `min-p` and that of the best other sampler. The conclusion is unchanged: `min-p` does not outperform other samplers when controlling for hyperparameter volume.

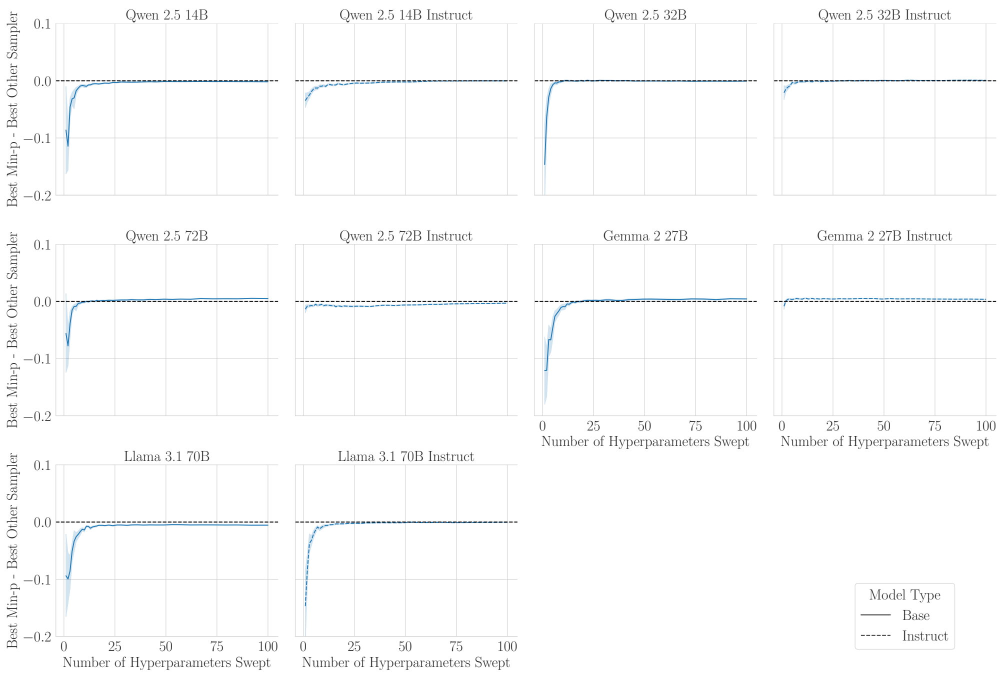

Figure 11: **`Min-P` Does Not Consistently Outperform Other Samplers on GSM8K CoT for Larger Models.** Difference between the best-of-N strict-match exact match of `min-p` and that of the best other sampler on GSM8K CoT (standard formatting) for ten models from 14B to 72B parameters.

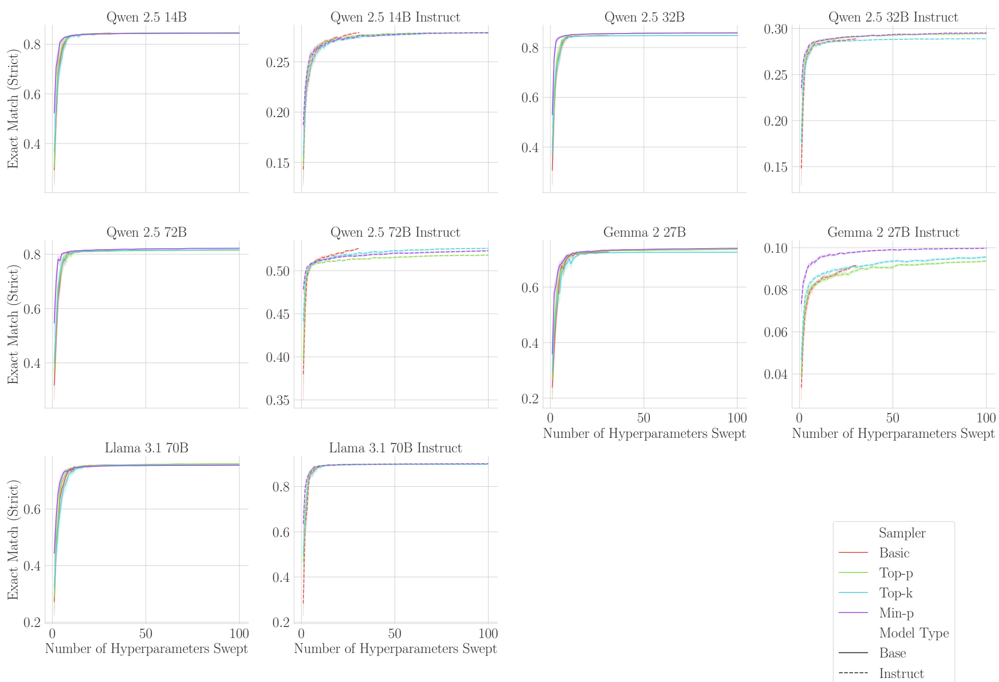

Figure 12: **Best-of-N GSM8K CoT Scores Per Sampler for Larger Models.** Companion to Fig. 11.

E The ICLR 2025 Review Record

For completeness, we note how the issues documented in the main text relate to the public ICLR 2025 review record for Nguyen et al. (2024). The community adoption figures that were later retracted (Sec. 5) were cited as evidence of impact in the meta-review: the Area Chair wrote that `min-p` "is simple and is already widely adopted by the community (as mentioned by [Reviewer] D38H, 'The usage of it in 54,000 Github repositories alone is very impressive')", and Reviewer fwNb wrote that `min-p` "has good empirical results, both as measured on benchmarks and (more important [sic]) by adoption of the community". The public reviews do not discuss which model(s) were sampled from in the LLM-as-a-judge evaluations or the absence of uncertainty estimates for the reported win rates (Sec. 4). Separately, the meta-review attributes `min-p`'s advantage to the low temperature regime ("in the low temperature regime, [`min-p`] provides a significant advantage"), whereas the paper's claims concern the high temperature regime.

F GSM8K Scores By Model, Sampler and Hyperparameters

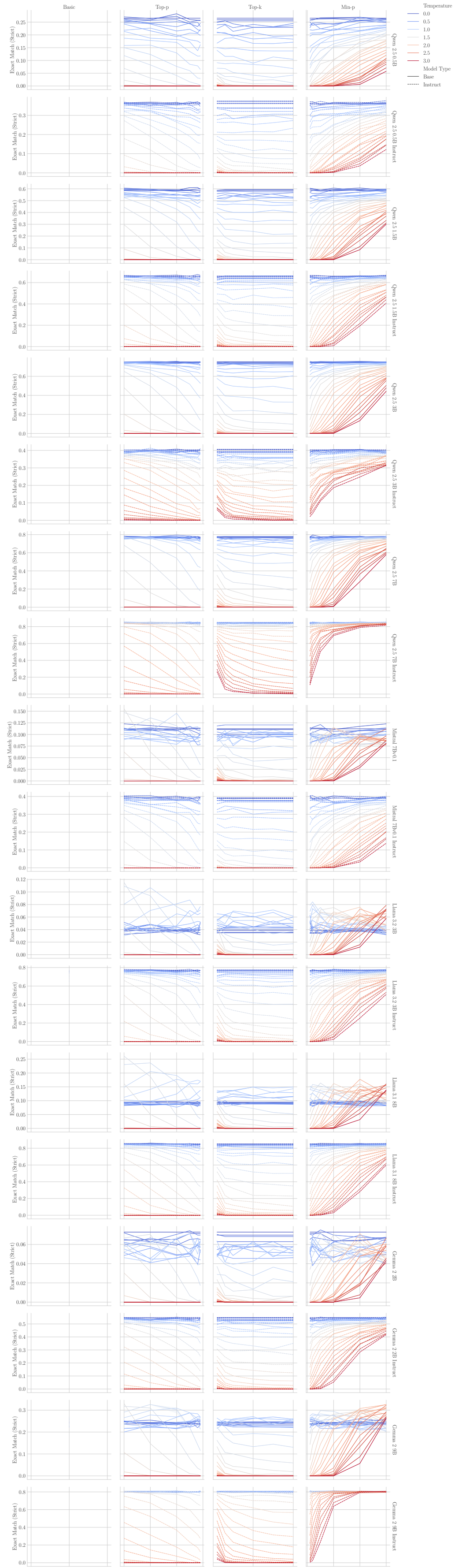

Figure 13: **GSM8K Scores By Model, Sampler and Sampler Hyperparameters.** Many models achieve their highest scores at low temperatures across samplers.